A Statistical Physics Approach to Market Dynamics: Exploring Phase Transitions in Digital vs. Local Vendor Choice

Abstract. Digital commerce platforms compete with local vendors under frictions of time, price, and visibility. We test whether an interpretable choice model with reinforcement and social influence can yield abrupt changes in aggregate market share. We simulate repeated customer choice among vendors using a utility function that combines vendor attributes (price, delivery or travel time, rating), reinforcement from past interactions (optionally decayed), and a prior-popularity social term. Choices follow a Boltzmann rule with temperature T controlling decision noise. An information parameter γ controls discovery of local options; $\gamma = 0$ keeps locals invisible and yields zero local purchases by construction. We measure late-time local share, market concentration (HHI), and diversity (entropy). In a baseline setting ($T = 0.5$, $\gamma = 0.3$, $\phi = 0.2$), the market is concentrated and local share is low (≈ 0.17). Enabling an Open Network for Digital Commerce (ONDC)-style protocol and increasing the protocol distance weight w increases local competitiveness at the same T and γ: late local share rises from ≈ 0.17 ($w = 0$) to ≈ 0.55 ($w = 1$), while HHI decreases from ≈ 0.64 to ≈ 0.30 and entropy increases from ≈ 0.79 to ≈ 1.44. A $\gamma \times w$ interaction slice shows that γ gates local participation (local share ≈ 0 for $\gamma = 0$ at all w), while w modulates outcomes once locals are discoverable ($\gamma > 0$), including at $\gamma = 1$. These results connect utility-based choice to macroscopic outcomes and isolate how information frictions and distance-weighted protocols affect platform–local competition.

Keywords: agent-based modeling; consumer choice; information frictions; market concentration; phase transitions

1 Introduction

1.1 Background and context

Digital commerce has fundamentally changed the way consumers shop and the way markets are organized. Large online platforms have grown rapidly by exploiting network effects—the tendency of a service to become more valuable as more people use it—which creates a self-reinforcing cycle where dominant sellers attract more buyers, and more buyers attract more sellers.[1] Research on two-sided market theory explains how these dynamics can lock entire markets into configurations where one or a few players capture a disproportionate share of all transactions, making it structurally difficult for newer or smaller sellers to compete.[2] A related phenomenon is technological lock-in: once consumers settle on a particular platform

or seller, switching costs and habit formation keep them there even when alternatives might be equally good.[3]

One group that is especially disadvantaged by these dynamics is local vendors—small, geographically-rooted sellers who may offer competitive prices, faster delivery for nearby customers, and direct community relationships, but who lack the algorithmic visibility and marketing budgets of large digital platforms. In many cases, customers may not know that a local vendor exists at all, even when that vendor would be a rational choice. This type of structural invisibility can produce what economists call information cascades: consumers continue buying from well-known sellers not because those sellers are objectively better, but simply because they have never been exposed to the alternatives.[4]

This paper approaches these dynamics through a physics-inspired lens. Researchers in the fields of econophysics and sociophysics have shown that the tools of statistical physics—originally developed to describe how atoms and molecules behave collectively—can also be applied to describe how large groups of people make decisions in markets and social systems.[5,6] A particularly powerful concept borrowed from physics is the *phase transition*: the idea that a system can shift abruptly from one qualitatively different state to another when a key control parameter crosses a threshold, much as water transitions sharply from liquid to ice at a specific temperature.[7,8] Markets can behave similarly. A market that begins in a diverse, balanced state can, under conditions of increasing social influence or growing information asymmetry, tip suddenly into a winner-take-most configuration.[9] The phase-transition analogy is valuable because it draws attention to a small number of interpretable control parameters—noise, information access, and social influence—that govern which regime the system occupies.

1.2 Problem statement and rationale

We ask: *How do randomness in individual choice and information frictions about local vendors shape the long-run market share of local commerce?* Two mechanisms are of particular interest. First, we model a temperature-like parameter T that controls how strongly customers follow utility differences versus exploring at random—a formalization grounded in the statistical mechanics of decision-making.[10,11] Second, we model an information-access rate γ that controls how quickly customers discover local vendors who were initially invisible to them. We also consider a social-influence term that biases customers toward vendors that were popular in the previous round, drawing on a long tradition of models in which social interactions generate multiple possible market equilibria depending on the strength of conformity.[12,13]

Understanding these mechanisms matters because information frictions are not simply a byproduct of market competition—they are often a structural feature that policy interventions can change. If altering information access can move a market from a concentrated,

digital-dominant regime to a more balanced one, then such interventions can be studied and evaluated with interpretable parameters rather than opaque prediction models.

1.3 Significance and purpose

These questions are directly relevant to real-world policy. The Open Network for Digital Commerce (ONDC) is a government-supported initiative in India, launched by the Department for Promotion of Industry and Internal Trade, that aims to democratize digital commerce by creating an open, interoperable protocol for buyer and seller applications.[14] In contrast to existing platforms like Amazon or Flipkart, where transactions are confined within a single proprietary ecosystem, ONDC establishes a shared technical standard that allows any buyer-side application to discover sellers listed on any seller-side application. The core idea is that by making local vendors discoverable across many platforms simultaneously, rather than invisible to all but one, the protocol reduces the information friction that keeps customers away from nearby sellers.

In the language of our model, ONDC-style interventions can be understood as operating on two levels simultaneously. First, they increase the effective discovery rate γ by making previously invisible local sellers findable. Second, when geographic proximity is weighted explicitly in search results or recommendations, they can increase the relative attractiveness of nearby local vendors through what we parameterize as a distance-weight w. The present study asks how large these effects need to be to meaningfully shift aggregate market outcomes.

If changes to information access or distance-weighted matching can move the market from a concentrated digital-dominant regime to a more balanced regime, then interventions of this kind can be evaluated using interpretable levers rather than black-box prediction models—a meaningful advantage for both policy design and public accountability.

1.4 Objectives

- Build an agent-based model of repeated customer–vendor choice in a mixed local/digital market.
- Quantify how late-time local market share and market concentration depend on key control parameters.
- Identify parameter regions where outcomes change rapidly (high sensitivity).
- Compare baseline dynamics with and without social influence ($\phi = 0$ ablation).

1.5 Scope and limitations

The model is intentionally stylized: vendors do not strategically change prices, customers use simplified utilities, and geography is represented as a unit square with Euclidean distance. Vendor attributes are exogenous random draws; the focus is on demand-side dynamics, specifically choice noise, learning, and information access. The goal is not to forecast the behavior of any specific market, but to test how a small set of mechanisms can generate regime-like shifts in aggregate outcomes under controlled conditions.

1.6 Theoretical framework

We track an order-parameter-like outcome, the *local share* $L \in [0, 1]$, defined as the fraction of all purchases going to local vendors in a given round. In physics, an order parameter is a quantity that is zero in one phase and nonzero in another—here, L serves as a macroscopic summary of which type of vendor the market favors. The main control parameters are the choice-noise scale T, the information-access rate γ, and the social-coupling strength ϕ. In this framing, L is a macroscopic observable whose dependence on control parameters can reveal regime boundaries.

We additionally track three summary statistics of market structure. The Herfindahl–Hirschman Index (HHI) is a standard measure of market concentration: it equals the sum of squared market shares across all vendors, ranging from near zero (very many small sellers) to one (single monopoly seller). Entropy, borrowed from information theory, measures diversity in the opposite direction: higher entropy means purchases are more evenly spread. Top-vendor share simply records the fraction of purchases captured by the single most popular vendor. Together, these three measures provide a multi-angle view of whether the market is becoming more or less concentrated as parameters change.

1.7 Methodology overview

We simulate N heterogeneous customers choosing among M vendors over many rounds. Each round updates (i) which local vendors each customer can see, (ii) choice probabilities derived from a utility model, and (iii) reinforcement from past purchases. We summarize outcomes using late-time local share, concentration (HHI), and diversity (entropy), and repeat runs to estimate average behavior under stochasticity.

2 Methods

2.1 Research design

We use an agent-based simulation of repeated purchase decisions in a stylized market with two vendor types: *local* (physically located) and *digital* (delivery-based). Agent-based modeling (ABM) is a computational approach in which individual agents—in this case, customers—are given simple behavioral rules, and the collective behavior that emerges from many agents interacting over time is then studied.[15,16] ABM is well-suited to questions like ours because it makes individual-level mechanisms explicit and visible, rather than assuming an aggregate equilibrium from the outset.

Each simulation run proceeds for a fixed number of rounds, and each parameter setting is repeated with different random seeds to average over stochasticity. We focus on two main control parameters: (i) a temperature-like noise scale T controlling how deterministic choices are, and (ii) an information-access rate γ controlling how quickly customers discover local vendors. We additionally consider social influence (ϕ) and an ONDC-style protocol switch (`protocol_enabled`) with a distance-weight parameter w (`protocol_distance_weight`). Digital vendors are always visible to all customers. Local vendors are discovered stochastically; $\gamma = 0$ implies local vendors remain completely invisible throughout the run and receive zero purchases by construction.

2.2 Participants or sample

Agents are synthetic. Each run contains N customers and M vendors placed uniformly at random in a unit-square geography (Appendix Fig. 4). Customers have heterogeneous preference parameters, including price sensitivity α_i and delay sensitivity u_i, drawn from fixed distributions with a negative correlation to represent a time–money trade-off: customers who are more price-sensitive tend to be less urgency-sensitive, and vice versa (Appendix Figs. 5–6). Vendors are assigned to be local or digital according to a fixed local fraction, and each vendor is assigned attributes such as price p_j, rating r_j, production cost c_j, and (for digital vendors) a delivery-delay parameter τ_j (Appendix Figs. 7–8).

2.3 Data collection

At each simulation round t, every customer chooses one vendor. We record per-round choice counts $\{n_j(t)\}_{j=1}^{M}$ and compute four summary time series:

$$L(t) = \sum_{j \in \mathcal{L}} \frac{n_j(t)}{N}, \tag{1}$$

$$S_{\max}(t) = \max_j \frac{n_j(t)}{N}, \tag{2}$$

$$\mathrm{HHI}(t) = \sum_{j=1}^{M} \left(\frac{n_j(t)}{N} \right)^2, \tag{3}$$

$$H(t) = -\sum_{j=1}^{M} \frac{n_j(t)}{N} \ln\left(\frac{n_j(t)}{N} \right). \tag{4}$$

with the convention that $0 \ln 0$ is treated as 0.

2.4 Variables and measurements

Choice temperature (T). Customer choices follow a Boltzmann (softmax) rule, a formalization of stochastic choice that originates in statistical mechanics and has been extensively applied in discrete choice economics.[17,18] Higher T produces more random exploration—customers pay less attention to utility differences—while lower T makes choices more deterministic, so small utility advantages reliably translate into larger market shares.

Information access (γ). Customers do not initially see every local vendor. Each customer maintains a visibility mask over local vendors. At each round, with probability γ, a customer discovers additional local vendors, with discovery biased toward geographically closer vendors. This setup captures the real-world observation that local sellers are systematically less discoverable than large digital platforms, and that exposure accumulates gradually through proximity, word-of-mouth, or search.

Social influence (ϕ). A social term increases the utility of vendors with higher popularity in the previous round. This models the tendency of consumers to be influenced by what others are buying—a phenomenon well documented in both economics and sociophysics.[11,13] Setting $\phi = 0$ removes social influence entirely and serves as an ablation condition to isolate its effect.

Reinforcement and memory. Customers carry an affinity toward vendors they have chosen before. Affinities can decay geometrically at a rate controlled by `AFFINITY_DECAY`;

repeated purchases increase affinity. This captures brand loyalty and habit formation in a simple but interpretable way.

ONDC-style protocol and distance weight (w). When `protocol_enabled=1`, an additional distance-weighted mechanism is applied through `protocol_distance_weight` $= w$. Conceptually, this represents the kind of proximity-aware discovery that an open digital commerce protocol could provide: nearby vendors are boosted in customers' effective consideration sets in proportion to w. In our experiments, w is varied in $[0, 1]$.

2.5 Procedure

Each simulation run follows the steps below.

1. **Initialize population.** Generate customers and vendors (positions and attributes) using a fixed random seed.

2. **Initialize information masks.** For each customer, mark all digital vendors as visible and initialize local-vendor visibility according to the run configuration.

3. **Iterate purchase rounds.** For $t = 1, \ldots, R$:

   (a) *Information update:* with probability γ, update the local-vendor visibility mask by adding newly discovered local vendors (distance-biased).

   (b) *Utility computation:* compute utilities $U_{ij}(t)$ for each visible customer–vendor pair (i, j).

   (c) *Stochastic choice:* convert utilities into choice probabilities via a softmax and sample one vendor for each customer.

   (d) *Learning update:* update customer affinities based on the chosen vendor and apply affinity decay (if enabled).

4. **Compute late-time outcomes.** Average $L(t)$, $\mathrm{HHI}(t)$, $H(t)$, and $S_{\max}(t)$ over a fixed late-time window to obtain "late" metrics.

Utility model. We use a linear utility function combining vendor attributes, customer-specific sensitivities, reinforcement, and social influence.[19,20] This type of model, sometimes called a random utility model, assumes that each customer chooses the vendor that provides the highest utility, where utility is the sum of observable attributes (price, rating, delay) and an idiosyncratic noise term. The specific form used here is:

$$U_{ij}(t) = -\alpha_i \, \tilde{p}_{ij} + \tilde{r}_j - u_i \, \tilde{\tau}_{ij} + \delta_i \, \log(1 + A_{ij}(t)) + \phi \, \log(1 + s_j(t-1)) + s_i \, \epsilon_{ij}(t), \qquad (5)$$

where $s_j(t-1) = n_j(t-1)/N$ is the previous-round share, $A_{ij}(t)$ is the affinity state, and $\epsilon_{ij}(t)$ is i.i.d. Gaussian noise scaled by s_i. Tildes indicate feature rescaling (e.g., standardization) so that T has a consistent interpretation across runs.

Delay term and geography. For local vendors, delay τ_{ij} is proportional to Euclidean distance between customer i and vendor j converted into minutes. This is the key channel through which geographic proximity enters the utility of local vendors—a nearby local seller is penalized less than a distant one, all else equal. For digital vendors, delay is based on the vendor's delivery parameter τ_j.

Choice rule. Given utilities over vendors visible to customer i, the probability of choosing vendor j is

$$P_{ij}(t) = \frac{\exp(U_{ij}(t)/T)}{\sum_{k \in \mathcal{V}_i(t)} \exp(U_{ik}(t)/T)}, \tag{6}$$

where $\mathcal{V}_i(t)$ is the current consideration set. This softmax rule, equivalent to the Boltzmann distribution in physics, is the standard formulation in the discrete choice literature for modeling stochastic preferences.[10,18]

2.6 Data analysis

For each parameter setting, we run multiple independent repeats and report mean late-time outcomes. Where applicable, we compute finite-difference sensitivities of late local share L:

$$\chi_T \approx \frac{L(T + \Delta T, \gamma) - L(T - \Delta T, \gamma)}{2\Delta T}, \qquad \chi_\gamma \approx \frac{L(T, \gamma + \Delta\gamma) - L(T, \gamma - \Delta\gamma)}{2\Delta\gamma}, \tag{7}$$

using one-sided differences at grid boundaries. Large values of χ_T or χ_γ at a particular operating point indicate that the system is in a sensitive region of parameter space—analogous to high susceptibility near a physical phase transition.

2.7 Ethical considerations

All agents and data are synthetic. No human participants, personal data, or identifiable information were collected.

3 Results

3.1 Baseline with ONDC protocol disabled

We first characterize the baseline model with the ONDC-style protocol turned off (`protocol_enabled`=0). Unless stated otherwise, all "late" metrics are averages over the final five rounds.

Run configuration (fixed across all repeats): $N = 200$ customers, $M = 40$ vendors, local_fraction = 0.5, $T = 0.5$, $\gamma = 0.3$, $\phi = 0.2$, rounds = 200, $\rho = 1.2$, $\sigma = 0.20$, affinity_decay = 1.0, protocol_distance_weight = 0.35. We ran 10 independent repeats.

Table 1 reports mean ± standard deviation across repeats. Figure 1 shows a representative equilibration trajectory: aggregate metrics settle quickly and then fluctuate weakly around a steady level. The high HHI (≈ 0.64) and top-vendor share (≈ 0.79) in this baseline confirm that without any discovery-enhancing intervention, the market is heavily concentrated in favor of a small number of vendors—in practice, the digital sellers who are always visible.

Table 1: ONDC-off baseline summary (mean ± s.d. over 10 runs). "Late" metrics are averages over the last five rounds.

Metric	Final	Late
Local share	0.172 ± 0.017	0.170 ± 0.017
HHI	0.642 ± 0.026	0.643 ± 0.024
Entropy	0.788 ± 0.055	0.785 ± 0.050
Top-vendor share	0.790 ± 0.019	0.791 ± 0.018

3.2 Protocol-enabled results (ONDC on)

We next enable the ONDC-style protocol (`protocol_enabled`=1) and vary the protocol distance-weight parameter $w \in [0, 1]$ (`protocol_distance_weight`). In this implementation, w controls how strongly geographic proximity is emphasized by the protocol mechanism.

Effect of distance weight w at fixed (T, γ, ϕ). At fixed $T = 0.5$, $\gamma = 0.3$, and $\phi = 0.2$, increasing w produces higher late-time local share and lower market concentration (HHI and top share), alongside higher entropy (Table 2). The $w = 0$ condition matches the ONDC-off baseline in Table 1, consistent with the protocol having no effect when its distance weighting is inactive. As w increases from 0 to 1, late local share more than triples (from ≈ 0.17 to ≈ 0.55), while HHI roughly halves and entropy nearly doubles—a substantial shift in market structure driven entirely by a change in how proximity is weighted in vendor recommendations.

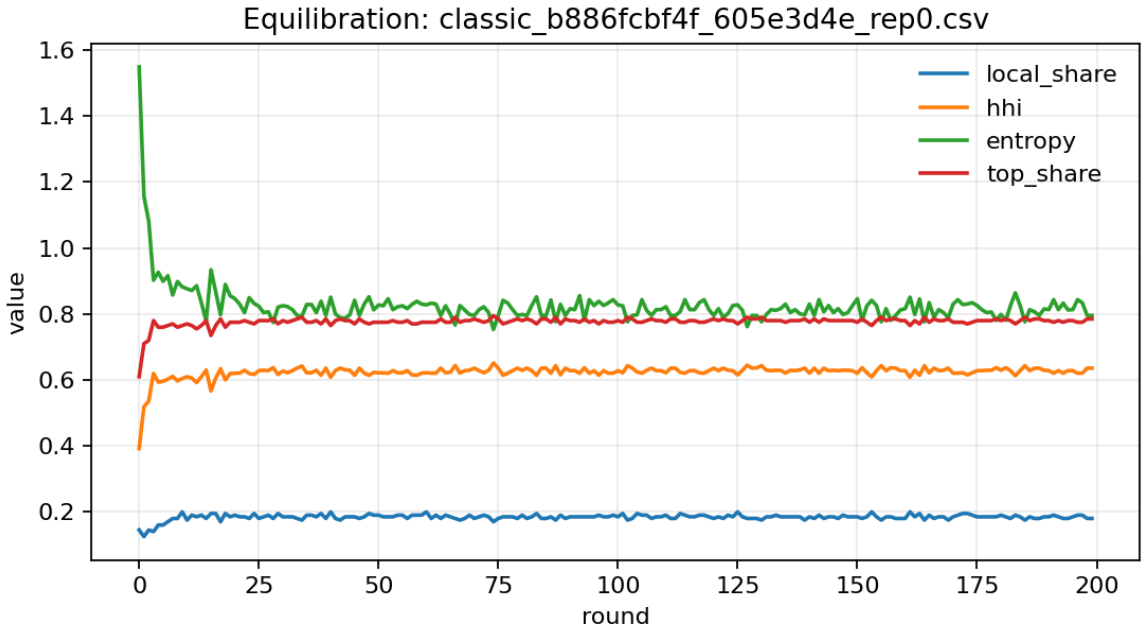


Figure 1: Representative equilibration time series for the ONDC-off baseline ($T = 0.5$, $\gamma = 0.3$, $\phi = 0.2$).

Table 2: ONDC-enabled distance-weight sweep at $T = 0.5$, $\gamma = 0.3$, $\phi = 0.2$ (mean ± s.d. over 10 runs).

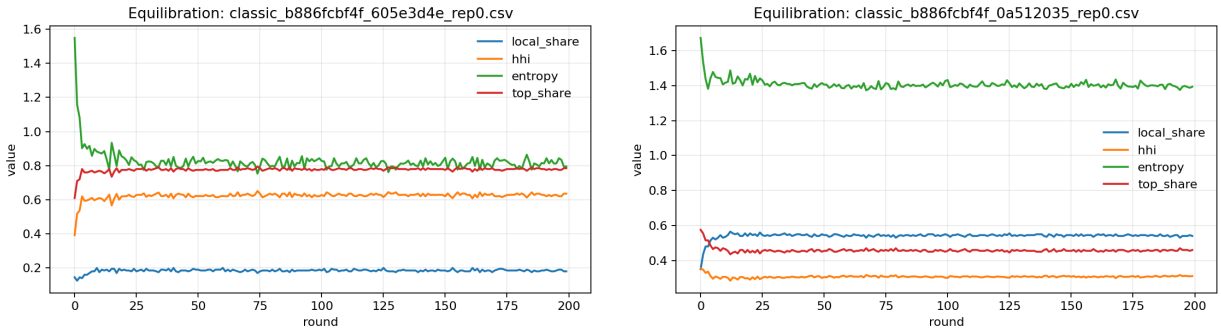
w	Late local share	Late HHI	Late entropy	Late top share
0.00	0.170 ± 0.017	0.643 ± 0.024	0.785 ± 0.050	0.791 ± 0.018
0.35	0.350 ± 0.028	0.465 ± 0.030	1.098 ± 0.055	0.648 ± 0.028
0.70	0.481 ± 0.021	0.347 ± 0.016	1.329 ± 0.036	0.518 ± 0.022
1.00	0.554 ± 0.019	0.299 ± 0.010	1.436 ± 0.028	0.445 ± 0.019

Representative comparison (ONDC off vs. ONDC on). Figure 2 contrasts a representative ONDC-off trajectory with a representative ONDC-on trajectory at high w. The ONDC-on setting shows higher local share and lower concentration relative to the ONDC-off baseline, with the difference visible from early rounds onward.

3.2.1 Interaction slice: information access γ × distance weight w at fixed T

To test whether ONDC primarily behaves like a change in local-vendor visibility, we varied the discovery rate γ together with the protocol distance weight w at fixed $T = 0.5$ and $\phi = 0.2$. The full grid of late local share values (mean ± s.d. across 10 runs per setting) is reported in Appendix Table 3.

Two descriptive patterns follow directly from the grid:



(a) ONDC off (`protocol_enabled`=0). (b) ONDC on (`protocol_enabled`=1, high w).

Figure 2: Representative equilibration trajectories with the ONDC-style protocol disabled vs. enabled.

- **Visibility gate (by construction):** when $\gamma = 0$, local vendors are never discovered and late local share is 0 for all w.
- **w-dependence once discoverable:** for $\gamma > 0$, late local share varies substantially with w, including at $\gamma = 1$.

Figure 3 provides one visualization of this interaction slice.

4 Discussion

Across these experiments, the information-access parameter γ functions as a strict gate for local participation: when $\gamma = 0$, local vendors remain invisible and receive zero purchases by construction, regardless of any protocol settings (Appendix Table 3). This result underscores a point that is easy to overlook in policy discussions—no amount of proximity weighting or protocol design can benefit local sellers who are never discovered in the first place. Visibility itself is a necessary precondition for competition, not just an advantage once competition is established.

For $\gamma > 0$, enabling the ONDC-style protocol and increasing the distance weight w is consistently associated with higher late local share and lower concentration, as reflected in Table 2. The magnitude of this effect is notable: moving from $w = 0$ to $w = 1$ at $\gamma = 0.3$ more than triples the local share, from approximately 0.17 to 0.55, while roughly halving the HHI from 0.64 to 0.30. This is a large change in market structure induced purely by shifting the weight placed on geographic proximity in the discovery mechanism, without any change to vendor prices, ratings, or other attributes.

The interaction slice further shows that w continues to modulate outcomes even under full discovery ($\gamma = 1$), indicating that the protocol effect is not simply about making locals visible—it also matters how strongly geographic proximity is weighted once locals are in

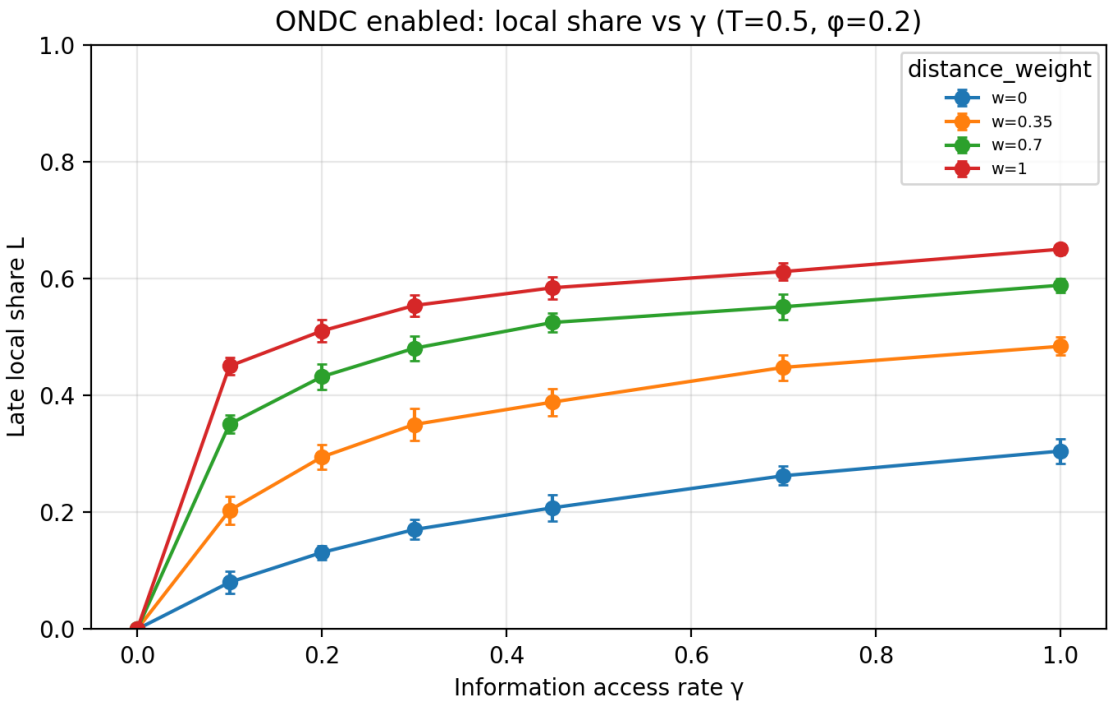


Figure 3: ONDC-on interaction slice at fixed $T = 0.5$ and $\phi = 0.2$. Late local share is shown across the $\gamma \times w$ sweep. Each curve corresponds to a different distance weight w; error bars show standard deviation across 10 runs.

the consideration set. In terms of the model's structure, this confirms that γ and w operate through distinct channels: γ determines whether a vendor enters the customer's consideration set at all, while w adjusts how attractive nearby vendors appear once they are there.

This study is limited by its stylized design: vendor attributes are fixed random draws, customers follow a simplified utility model, and vendor-side strategy (e.g., pricing responses to new competition) is not included. These are meaningful simplifications. In a real market, vendors would adjust their prices and marketing in response to changing competition, and customers would have richer preference structures than a linear utility function can capture. Within these constraints, however, the results illustrate how interpretable mechanisms for information access and distance-weighting can correspond to large, consistent changes in aggregate market outcomes. The framework is not meant to predict any specific market precisely, but to isolate and study the mechanisms that real policy interventions engage.

Future work can extend the same framework in several directions: expanding the parameter sweeps to include a broader range of T, testing additional ablations or model variations already described in the Methods (such as varying social influence strength or affinity decay), and incorporating simple forms of vendor-side strategic response. Such extensions would allow a richer exploration of the conditions under which distance-weighted discovery protocols produce robust versus fragile improvements in local-vendor market access.

A Appendix: Population and attribute generation

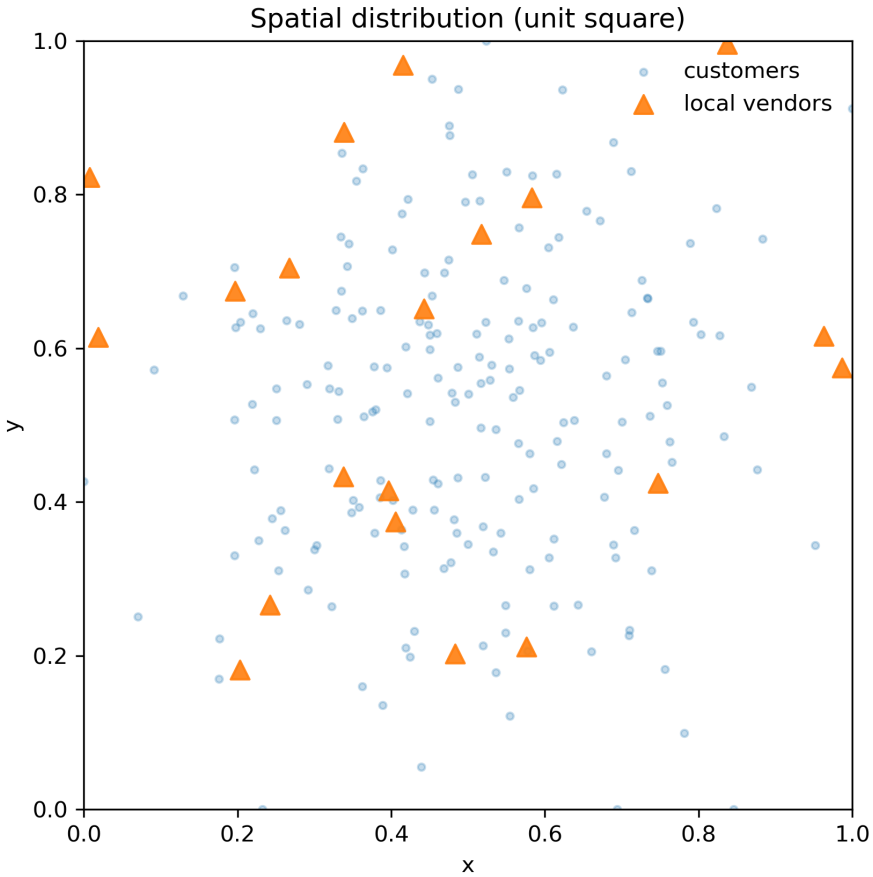


Figure 4: Spatial distribution of customers (points) and local vendors (triangles) in the unit-square geography.

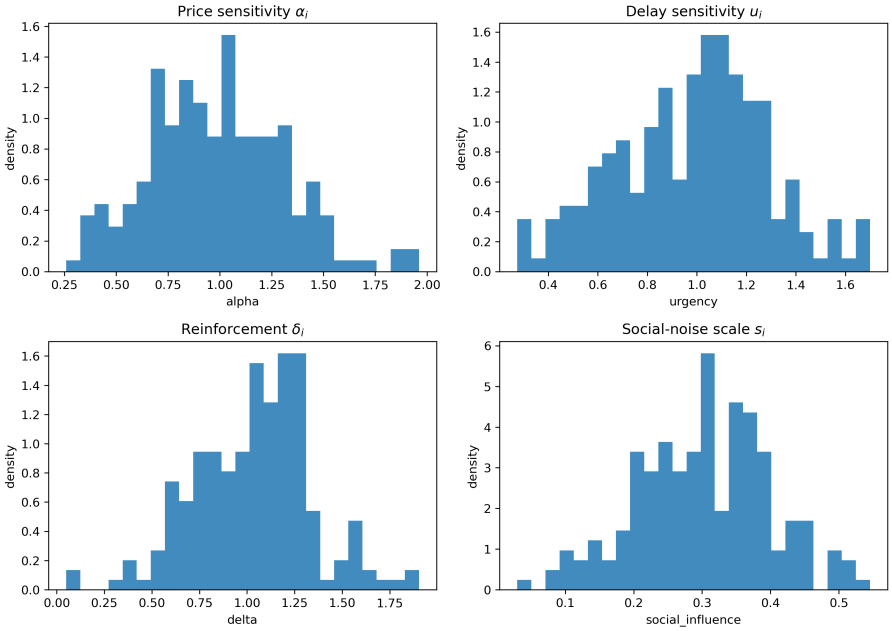


Figure 5: Customer heterogeneity distributions (price sensitivity, delay sensitivity, reinforcement strength, and social-noise scale).

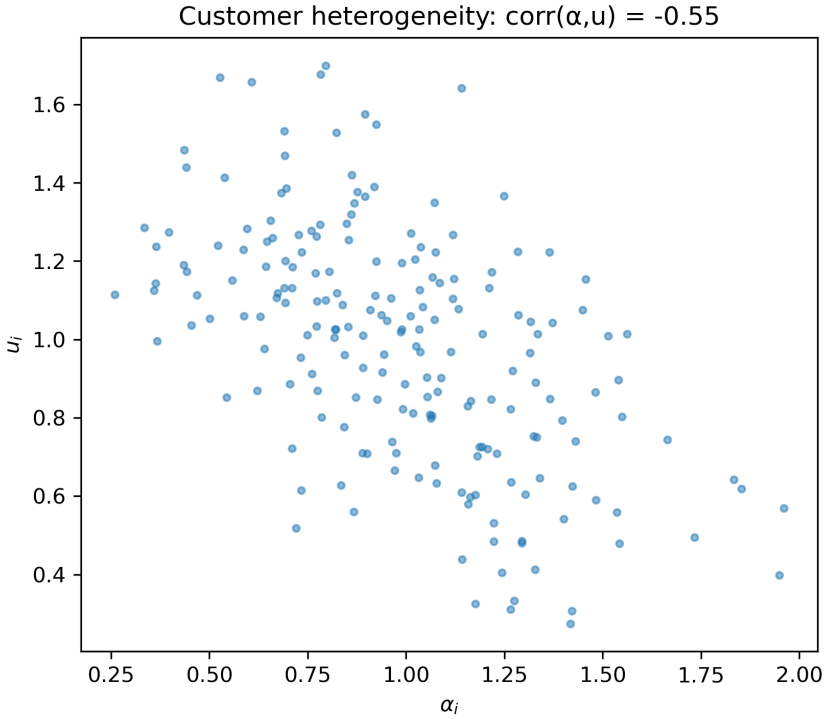


Figure 6: Joint distribution of (α_i, u_i) showing a negative correlation (time–money trade-off).

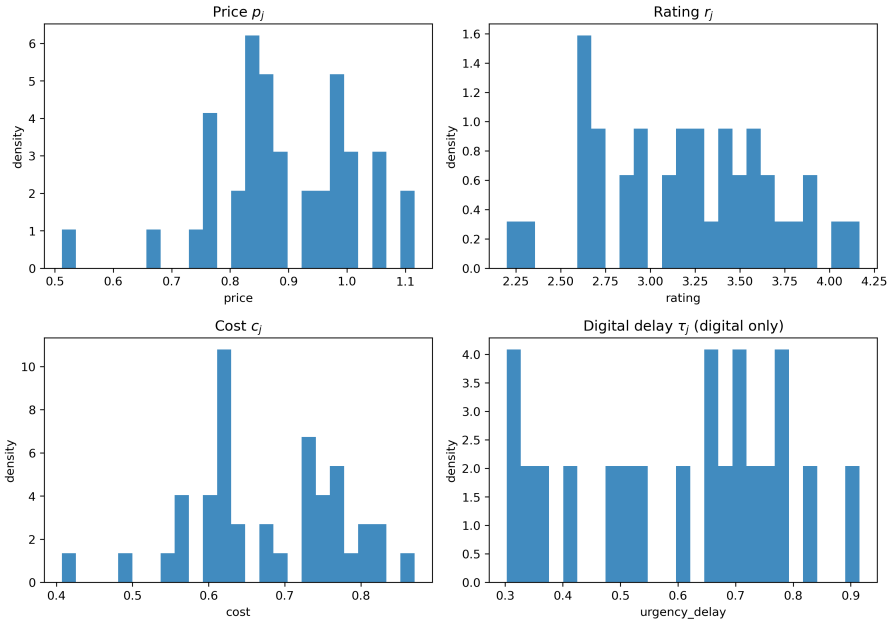


Figure 7: Vendor attribute distributions (price, rating, cost, and a delivery-delay parameter for digital vendors).

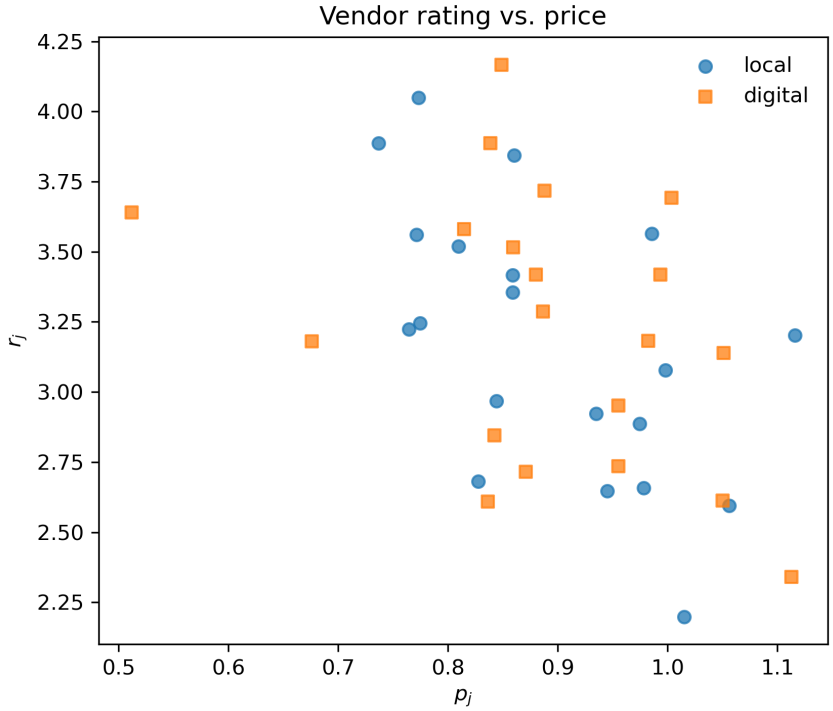


Figure 8: Example scatter of vendor rating versus price, separated by local vs. digital type.

B Appendix: Full ONDC interaction slice

Table 3: Late local share (mean ± s.d., 10 runs) for the ONDC-enabled interaction slice at $T = 0.5$, $\phi = 0.2$. Columns correspond to ONDC distance weight w.

γ	$w = 0.0$	$w = 0.35$	$w = 0.7$	$w = 1.0$
0.00	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
0.10	0.142 ± 0.013	0.280 ± 0.029	0.374 ± 0.017	0.428 ± 0.023
0.20	0.156 ± 0.014	0.305 ± 0.023	0.409 ± 0.013	0.472 ± 0.023
0.30	0.170 ± 0.017	0.350 ± 0.028	0.481 ± 0.021	0.554 ± 0.019
0.45	0.201 ± 0.015	0.415 ± 0.032	0.571 ± 0.025	0.645 ± 0.027
0.70	0.256 ± 0.021	0.505 ± 0.036	0.635 ± 0.029	0.691 ± 0.023
1.00	0.301 ± 0.018	0.589 ± 0.030	0.689 ± 0.020	0.653 ± 0.019

References

[1] M. L. Katz, C. Shapiro. Network externalities, competition, and compatibility. American Economic Review. Vol. 75, pg. 424–440, 1985.

[2] J.-C. Rochet, J. Tirole. Platform competition in two-sided markets. Journal of the European Economic Association. Vol. 1, pg. 990–1029, 2003, https://doi.org/10.1162/154247603322493212.

[3] W. B. Arthur. Competing technologies, increasing returns, and lock-in by historical events. Economic Journal. Vol. 99, pg. 116–131, 1989, https://doi.org/10.2307/2234208.

[4] S. Bikhchandani, D. Hirshleifer, I. Welch. A theory of fads, fashion, custom, and cultural change as informational cascades. Journal of Political Economy. Vol. 100, pg. 992–1026, 1992, https://doi.org/10.1086/261849.

[5] C. Castellano, S. Fortunato, V. Loreto. Statistical physics of social dynamics. Reviews of Modern Physics. Vol. 81, pg. 591–646, 2009, https://doi.org/10.1103/RevModPhys.81.591.

[6] R. N. Mantegna, H. E. Stanley. *An introduction to econophysics: Correlations and complexity in finance*. Cambridge University Press, 2000.

[7] H. E. Stanley. *Introduction to phase transitions and critical phenomena*. Oxford University Press, 1971.

[8] D. Sornette. *Critical phenomena in natural sciences: Chaos, fractals, selforganization and disorder*. 2nd ed. Springer, 2006.

[9] T. C. Schelling. *Micromotives and macrobehavior*. Norton, 1978.

[10] L. E. Blume. The statistical mechanics of strategic interaction. Games and Economic Behavior. Vol. 5, pg. 387–424, 1993, `https://doi.org/10.1006/game.1993.1023`.

[11] W. A. Brock, S. N. Durlauf. Discrete choice with social interactions. Review of Economic Studies. Vol. 68, pg. 235–260, 2001, `https://doi.org/10.1111/1467-937X.00168`.

[12] S. N. Durlauf. Nonergodic economic growth. Review of Economic Studies. Vol. 60, pg. 349–366, 1993, `https://doi.org/10.2307/2298061`.

[13] S. Galam, S. Moscovici. Towards a theory of collective phenomena: Consensus and attitude changes in groups. European Journal of Social Psychology. Vol. 21, pg. 49–74, 1991, `https://doi.org/10.1002/ejsp.2420210105`.

[14] Department for Promotion of Industry and Internal Trade. Open network for digital commerce: Policy and technical framework. Government of India, New Delhi, 2022. `https://ondc.org`.

[15] J. M. Epstein, R. Axtell. *Growing artificial societies: Social science from the bottom up*. MIT Press, 1996.

[16] L. Tesfatsion, K. L. Judd, eds. *Handbook of computational economics, volume 2: Agent-based computational economics*. Elsevier, 2006.

[17] K. Huang. *Statistical mechanics*. 2nd ed. Wiley, 1987.

[18] D. McFadden. Conditional logit analysis of qualitative choice behavior. In P. Zarembka, ed., *Frontiers in econometrics*. Academic Press, pg. 105–142, 1974.

[19] K. E. Train. *Discrete choice methods with simulation*. Cambridge University Press, 2003.

[20] S. P. Anderson, A. de Palma, J.-F. Thisse. *Discrete choice theory of product differentiation*. MIT Press, 1992.